AP CALCULUS AB

Unit 1 • Lesson 1-6 • Guided notes

Lesson 1-6 • Average Rate of Change

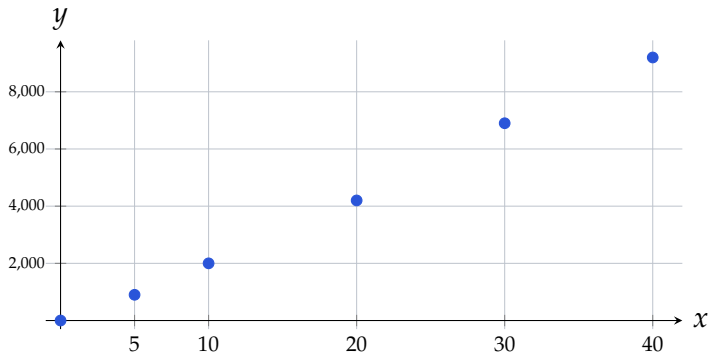
Guided notes • Fill in the blanks and work the examples in the boxes

Objective: find and interpret the average rate of change on an interval, with units.

Moussa's bike ride

Moussa rides his bike along the Corniche in Dakar. The distance he has ridden, in meters, t minutes after he starts, is $D(t)$ for $0 \le t \le 40$.

t (min)	0	5	10	20	30	40
$D(t)$ (m)	0	900	2000	4200	6900	9200



Average rate of change

The average rate of change of f on $a \le x \le b$: _____ $= \dfrac{\Delta y}{\Delta x}$

Graphically, it is the _____ of the _____ through the two points $(a, f(a))$ and $(b, f(b))$.

Units: _____.

Example 1

a) Sketch the line that represents the average rate of change of D over the whole interval.

b) Find the average rate of change of D on $0 \le t \le 40$. Give units.

Example 2

a) Sketch the line that represents the average rate of change of D on $5 \le t \le 20$.

b) Find it and interpret it in context.

Example 3	Find the average rate of change of $f(x) = x^2 - 2x$ on $[1, 4]$.
Summary	